打开 ida 分析下断点

```
 9   char v10[13]; // [rsp+53h] [rbp-2Dh] BYREF
10   int v11[100]; // [rsp+60h] [rbp-20h] BYREF
11   char v12[52][10]; // [rsp+1F0h] [rbp+170h] BYREF
12   int j; // [rsp+3F8h] [rbp+378h]
13   int i; // [rsp+3FCh] [rbp+37Ch]
14
15   _main();
16   uR3(v12);
17   qR2(v12, (int (*)[10])v11);
18   for ( i = 0; i <= 9; ++i )
19   {
20     for ( j = 0; j <= 9; ++j )
21     {
22       if ( !v11[10 * i + j] )
23         ++vX3;
24     }
25   }
26   dM7(0, 0, 0, (int (*)[10])v11);
27   if ( vZ0 )
28   {
29     xP9((int (*)[10])v11);
30     mS2(v10);
31     sprintf_s<19ull>(v9, "ISCC{%s}", v10);
32     ((void (__fastcall *)(char *))std::string::basic_string)(v8);
33     v5 = std::operator<<<std::char_traits<char>>(refptr__ZSt4cout, "Enter flag:");
34     ((void (__fastcall *)(__int64))refptr__ZSt4endlIcSt11char_traitsIcEERSt13basic_ostreamIT_T0_ES6_)(v5);
35     std::getline<char,std::char_traits<char>,std::allocator<char>>(refptr__ZSt3cin, v8);
36     if ( (unsigned __int8)std::operator==<char>(v8, v9) )
37       v6 = std::operator<<<std::char_traits<char>>(refptr__ZSt4cout, "Correct!");
38     else
39       v6 = std::operator<<<std::char_traits<char>>(refptr__ZSt4cout, "Incorrect!");
40     ((void (__fastcall *)(__int64))refptr__ZSt4endlIcSt11char_traitsIcEERSt13basic_ostreamIT_T0_ES6_)(v6);
41     system("pause");
42     v4 = 0;
43     ((void (__fastcall *)(char *))std::string::~string)(v8);
44   }
45   else
46   {
47     v3 = std::operator<<<std::char_traits<char>>(refptr__ZSt4cout, "Error: no path!");
48     ((void (__fastcall *)(__int64))refptr__ZSt4endlIcSt11char_traitsIcEERSt13basic_ostreamIT_T0_ES6_)(v3);
49     return 1;
50   }
51   return v4;
52 }
000013EB main:31 (401DEB)
```

Dbg 一下

```
.text:0000000000401DBE test    eax, eax
.text:0000000000401D90 jnz     short loc_401DBE

.text:0000000000401DBE
.text:0000000000401DBE loc_401DBE:
.text:0000000000401DBE lea     rax, [rbp+390h+var_3B0]
.text:0000000000401DC2 mov     rcx, rax        ; int (*)[10]
.text:0000000000401DC5 call    _Z3xP9PA10_i    ; xP9(int (*)[10])
.text:0000000000401DCA lea     rax, [rbp+390h+var_3BD]
.text:0000000000401DCE mov     rcx, rax        ; char *
.text:0000000000401DD1 call    _Z3mS2Pc        ; mS2(char *)
.text:0000000000401DD6 lea     rdx, [rbp+390h+var_3BD]
.text:0000000000401DDA lea     rax, [rbp+390h+var_3D0]
.text:0000000000401DDE mov     r8, rdx
.text:0000000000401DE1 lea     rdx, aIsccS     ; "ISCC{%s}"
.text:0000000000401DE8 mov     rcx, rax
.text:0000000000401DEB call    _Z9sprintf_sILy19EEiRAT__cPKcz ; sprintf_s<19ull>(char (&)[19ull],char const*,...)
.text:0000000000401DF0 lea     rax, [rbp+390h+var_3F0]
.text:0000000000401DF4 mov     rcx, rax
.text:0000000000401DF7 call    _ZNSt7__cxx1112basic_stringIcSt11char_traitsIcESaIcEEC1Ev ; std::string::basic_string(void)
.text:0000000000401DFC lea     rdx, aEnterFlag ; "Enter flag:"
.text:0000000000401E03 mov     rcx, cs:_refptr__ZSt4cout
.text:0000000000401E0A ;   try {
.text:0000000000401E0A call    _ZStlsISt11char_traitsIcEERSt13basic_ostreamIcT_ES5_PKc ; std::operator<<<std::char_traits<char>>(std::ostream &,char const*)
.text:0000000000401E0F mov     rdx, cs:_refptr__ZSt4endlIcSt11char_traitsIcEERSt13basic_ostreamIT_T0_ES6_
100.00% (829,773) (200,167) 000013D6 0000000000401DD6: main+113 (Synchronized with RIP)

Hex View-1
0000000000401C80  CC C7 45 D8 00 00 00 00  83 7D D8 09 7F 2E 8B 45  ....E.....}.....E
0000000000401C90  D8 83 C0 1E 48 98 48 8D  15 C3 43 0B 00 0F B6 04  ....H.H...C.....
0000000000401CA0  10 83 F0 AA 89 C1 48 8B  45 10 48 8D 50 5A 8B 45  ......H.E.H.PZ..E
0000000000401CB0  D8 48 98 88 0C 02 83 45  D8 01 EB CC 90 48 83 C4  .H.....E.....H..
0000000000401CC0  30 5D C3 55 53 48 81 EC  08 04 00 00 48 8D AC 24  0].USH......H..$
0000000000401CD0  80 00 00 00 E8 37 AA 00  00 48 8D 85 70 01 00 00  .....7...H..p...
0000000000401CE0  48 89 C1 E8 8A FD FF FF  48 8D 55 E0 48 8D 85 70  H.......H.U.H..p
0000000000401CF0  01 00 00 48 89 C1 E8 65  F8 FF FF C7 85 7C 03 00  ...H...e.....|..
0000000000401D00  00 00 00 00 00 83 BD 7C  03 00 00 09 7F 5E C7 85  .......|.....^..
000012C0 0000000000401CC0: uR3(char (*)[10])+24E
```

F5 找到 v10

```
Debug View
IDA View-RIP
16  uR3((char (*)[10])v12);
17  qR2((const char (*)[10])v12, (int (*)[10])v11);
18  for ( i = 0; i <= 9; ++i )
19  {
20    for ( j = 0; j <= 9; ++j )
21    {
22      if ( !v11[10 * i + j] )
23        ++vX3;
24    }
25  }
26  dM7(0, 0, 0, (int (*)[10])v11);
27  if ( vZ0 )
28  {
29    xP9((int (*)[10])v11);
30    mS2(v10);
31    sprintf_s<19ull>(v9, "TSCC{%s}", v10);
32    std::string::basic_string(v8);
33    v5 = std::operator<<<std::char_traits<char>>(refptr__ZSt4cout, "Enter flag:");
34    ((void (__fastcall *)(__int64))refptr__ZSt4endlIcSt11char_traitsIcEERSt13basic_ostreamIT_T0_ES6_)(v5);
35    std::getline<char,std::char_traits<char>,std::allocator<char>>(refptr__ZSt3cin, v8);
36    if ( (unsigned __int8)std::operator==<char>(v8, v9) )
37      v6 = std::operator<<<std::char_traits<char>>(refptr__ZSt4cout, "Correct!");
38    else
39      v6 = std::operator<<<std::char_traits<char>>(refptr__ZSt4cout, "Incorrect!");
40    ((void (__fastcall *)(__int64))refptr__ZSt4endlIcSt11char_traitsIcEERSt13basic_ostreamIT_T0_ES6_)(v6);
41    system("pause");
42    v4 = 0;
43    std::string::~string(v8);
  }
000013D6 main:31 (401DD6)
Hex View-1
```

双击打开 v10 出 flag

```
Debug View
IDA View-RIP
Stack[0004CCF0]:000000000079FA5F db   0
Stack[0004CCF0]:000000000079FA60 db  59h ; Y
Stack[0004CCF0]:000000000079FA61 db   4
Stack[0004CCF0]:000000000079FA62 db   0
Stack[0004CCF0]:000000000079FA63 db  28h ; (
Stack[0004CCF0]:000000000079FA64 db  64h ; d
Stack[0004CCF0]:000000000079FA65 db  26h ; &
Stack[0004CCF0]:000000000079FA66 db  22h ; "
Stack[0004CCF0]:000000000079FA67 db  52h ; R
Stack[0004CCF0]:000000000079FA68 db  5Ch ; \
Stack[0004CCF0]:000000000079FA69 db  75h ; u
Stack[0004CCF0]:000000000079FA6A db  4Fh ; O
Stack[0004CCF0]:000000000079FA6B db  4Ah ; J
Stack[0004CCF0]:000000000079FA6C db  53h ; S
Stack[0004CCF0]:000000000079FA6D db  3Eh ; >
Stack[0004CCF0]:000000000079FA6E db  6Bh ; k
Stack[0004CCF0]:000000000079FA6F db   0
Stack[0004CCF0]:000000000079FA70 db   0
Stack[0004CCF0]:000000000079FA71 db   0
Stack[0004CCF0]:000000000079FA72 db   0
Stack[0004CCF0]:000000000079FA73 db   0
Stack[0004CCF0]:000000000079FA74 db   1
Stack[0004CCF0]:000000000079FA75 db   0
Stack[0004CCF0]:000000000079FA76 db   0
Stack[0004CCF0]:000000000079FA77 db   0
Stack[0004CCF0]:000000000079FA78 db   1
Stack[0004CCF0]:000000000079FA79 db   0
Stack[0004CCF0]:000000000079FA7A db   0
Stack[0004CCF0]:000000000079FA7B db   0
UNKNOWN 000000000079FA63: Stack[0004CCF0]:000000000079FA63 (Synchronized w
```